Partner-Specific Affective Precision in Social Active Inference

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Abstract. In multi-agent social settings, model reliability varies across relationships. Beyond inferring what others will do, an agent must calibrate how confidently those inferences should guide policy selection for each relationship. An agent may maintain a well-validated model of one partner, a fragile model of another, and a model under revision for a third; collapsing these into a single confidence estimate loses information critical for policy selection. Thus, we use active inference to model affect as a metacognitive estimate of confidence in the current partner model, formalized as relationship-specific affective precision. Each partner's behavioral evidence updates a local confidence estimate that modulates policy precision during selection, regulating how strongly inferred beliefs are expressed in policy rather than changing the beliefs themselves.
Simulations in a multi-partner graded trust game show that partner-local affective precision influences behavior through policy commitment rather than improved inference about partner states. Because the mechanism tracks partner-response predictability rather than realized payoff, greater confidence leads to more committed policies without necessarily producing higher rewards. Under abrupt shifts in social behavior, confidence accumulated from a previously reliable relationship can continue to sharpen commitment after the partner changes, showing that adaptation requires revising confidence estimates and reallocating engagement across relationships. Finally, varying precision gain and priors on model fitness reveal distinct trust-calibration profiles, generating computational hypotheses about individual differences in social trust. Together, these results demonstrate that affective precision serves as a relationship-specific metacognitive signal that distinguishes social prediction from social commitment.

Keywords: Active inference · Affective precision · Social metacognition · Predictive reliability · Policy precision · Trust calibration

1 Introduction

Social agents rarely act on a single, stable model of the social world; instead, they maintain different expectations for different partners. One partner may be familiar and reliable, another newly encountered and uncertain, and another predictable but adversarial. Repeated social exchange depends on learning such partner-specific regularities, including reciprocity, reputation, moral character, and social value [17, 3, 7]. In such settings, the question is not only what an agent believes about each partner, but how confidently those beliefs should guide policy inference.

Computational models of social decision-making can therefore be separated into three problems. The first is inferring a partner's behavioral disposition: whether their actions are cooperative, exploitative, reciprocal, or volatile. The second is modeling what they know or intend: the theory-of-mind problem of how another agent represents the world and plans action [11, 12, 21]. The third is estimating whether the resulting partner model is reliable enough to guide confident policy commitment. To address this third problem, an agent must decide not only what it believes about a partner, but how strongly to act on those beliefs. However, many formal models specify the content of social inference while treating confidence in those inferences as fixed, global, or implicit.

The active inference framework [9, 10, 19] provides a natural formalism for this problem because it links perception, belief updating, and policy selection through a common inferential process. In active inference, agents infer hidden states of the world and select policies by minimizing expected free energy. Moreover, policy inference is precision-weighted, controlling how sharply an agent commits to its inferred policy distribution, making it an effective model of not only what an agent believes but how confidently those beliefs are expressed in policy.

Existing social active-inference models address complementary parts of the problem. Theory-of-mind models address how agents model what others know or intend through recursive belief structures and factorized generative models [21, 22]. Volatility and social-learning models address how prediction errors should change belief updating under uncertainty [18, 3, 8]. Social-learning work also shows that agents revise expectations from discrepancies between expected and observed behavior of others, including in trust-game interactions [16]. These approaches clarify how agents infer partners, model others' mental states, and revise beliefs under uncertainty. However, they do not by themselves explain how confidence in a social model should remain local to the relationship that generated the evidence.

Affect provides a natural candidate mechanism for this confidence-calibration role. In psychology and neuroscience, affect is often described in terms of valence and arousal—the felt pull toward or away from situations and social commitments [2]. Free-energy and active-inference accounts extend this view by linking affective dynamics to prediction error, model evidence, emotional-state inference, and precision-weighted policy inference [15, 23, 14, 20]. Prior affective-precision accounts make confidence endogenous by relating affect to subjective model fitness and policy precision [14]. On this view, affect can regulate how strongly current beliefs are expressed in policy, shaping the degree of commitment rather than the content of inference.

We therefore define partner-local affective precision as a relationship-specific confidence signal. The focal agent maintains a separate β_k estimate for each partner, updated only from that partner's behavioral evidence, which modulates policy precision when inferring policies toward that partner. Unlike prior affective-precision accounts centered on policy-inference quantities [14], we formulate the affective signal from partner-response likelihood surprisal: the surprise of an observed partner response under the focal agent's current posterior over that partner's type and stance. This makes the signal perceptual and relationship-local: it tracks whether the current model predicts this partner's behavior, preserving predictability differences that a global confidence estimate would collapse.

We evaluate this mechanism in a repeated multi-partner graded trust game with partner-choice conditions, abrupt betrayal, and profile variation. The remainder of this paper is organized as follows: Section 2 specifies the multi-partner trust game, per-partner generative models, and the partner-local affective precision mechanism. Section 3 reports simulation results and analyses. Section 4 discusses implications, limitations, and directions for future work.

2 Methods

A focal active-inference agent plays a repeated multi-partner trust/investment game with four partners whose behavioral types and stances are hidden. The agent is implemented using the `pymdp` library for discrete active inference [13, 6], which provides standard components for belief updating and policy selection [9, 10, 19]. Partner-local affective precision is layered on top of separate per-partner generative models, allowing both evidence and policy precision to remain relationship-specific. Full POMDP specifications, affective-precision updates, and simulation protocols are provided in Appendices A–C.

2.1 Trust game

We evaluate the mechanism in a repeated multi-partner trust/investment game [4]. The task has a Prisoner's-Dilemma-style social-dilemma structure: investing more can yield larger returns when a partner cooperates, but larger losses relative to the risk-free baseline when the partner defects. The task therefore requires the focal agent to solve two coupled problems: predicting how each partner is likely to respond, and deciding how strongly to commit to that prediction through investment.

Each interaction consists of partner assignment or selection, investment, and response. In partner-choice conditions, the focal agent first selects a partner k; in conditions without explicit partner choice, the interaction partner is assigned according to the configured protocol. The agent then chooses a graded investment level $a \in \mathcal{A} = \{0, 1, 2, 3, 4, 5\}$, where 0 denotes no investment and 5 denotes maximum investment. The selected partner emits a response $o_k^{\mathrm{act}} \in \{\mathrm{cooperate}, \mathrm{defect}\}$, sampled from that partner's hidden behavioral state.

Each partner has two hidden state factors. The first is behavioral type, $s_k^{\mathrm{type}} \in \{\mathrm{cooperator}, \mathrm{reciprocator}, \mathrm{exploiter}, \mathrm{random}\}$. The second is stance toward the focal agent, $s_k^{\mathrm{stance}} \in \{\mathrm{trusting}, \mathrm{neutral}, \mathrm{hostile}\}$. These states are not directly observed; the focal agent infers them from partner responses and payoff observations. Partners themselves are environment-side processes: they sample cooperate-or-defect responses from type- and stance-conditioned probabilities and undergo investment-dependent stance transitions, but do not perform variational inference or maintain affective precision estimates. This isolates the focal agent's precision mechanism before extending to reciprocal active-inference partners.

With endowment $E = 10$ and return multiplier $M = 3$, focal payoff is

$$\pi(a, o_k^{\text{act}}) = \begin{cases} E - a + \frac{Ma}{2}, & o_k^{\text{act}} = \text{cooperate}, \\ E - a, & o_k^{\text{act}} = \text{defect}. \end{cases}$$

Thus, cooperation yields $\pi(a, \text{cooperate}) = 10 + 0.5a$, whereas defection yields $\pi(a, \text{defect}) = 10 - a$. The no-investment action $a = 0$ gives a risk-free payoff of 10, while higher investment increases both potential gains and exposure to defection.

Investment also affects future partner stance. Higher investment provides stronger cooperative evidence and shifts stance dynamics toward trust-building transitions; lower investment or withholding shifts dynamics toward trust-damaging transitions. Formally, the transition model interpolates between high-investment and low-investment endpoint matrices according to $\rho(a) = a/5$, with full transition matrices given in Appendix A. Graded actions are used as the primary evaluation regime because they allow partner-specific precision to modulate the strength of commitment, not merely the direction of a binary cooperate/defect choice.

2.2 Per-partner generative models

Rather than maintaining a single pooled social model, the focal agent maintains a separate discrete generative model $\mathcal{M}_k$ for each partner k. This factorization keeps observations, beliefs, and confidence evidence local to the relationship that generated them. Each $\mathcal{M}_k$ uses the standard $A/B/C/D/E$ components of discrete active inference [6]: observation likelihoods, transition dynamics, preferences, initial priors, and policy priors. Full matrix specifications are provided in Appendix A.

Each partner model contains three hidden factors: partner type, partner stance, and own investment. Type and stance are inferred from evidence; own investment is deterministic bookkeeping that makes the executed investment level available to the payoff likelihood. Observations are the partner's cooperate-or-defect response and the focal agent's realized payoff. The partner-response likelihood depends on type and stance, so predictive reliability is defined by response model fit rather than payoff: a reliably cooperative and a reliably defecting partner can both be predictable. The payoff likelihood depends on investment level and marginalizes over partner response, keeping instrumental payoff preferences inside the generative model. Transition dynamics are specified in Section 2.1 and Appendix A. This model supplies ordinary partner-state inference; the affective-precision layer in Section 2.3 estimates how confidently each partner model should be deployed in policy inference.

2.3 Partner-local affective precision

Partner-local affective precision maps relationship-specific model fit onto policy precision. After each interaction with partner k, the focal agent evaluates how surprising the observed partner response was under its current posterior over that partner's type and stance. Unlike prior affective-precision accounts centered on policy-inference quantities [14], the evidence source here is perceptual partner-response model fit. Because this update is computed separately for each partner, confidence can diverge across relationships even when the focal agent's task and preferences remain fixed.

Let $\hat{o}_k^{\text{act}}$ denote the observed response from partner k. The agent computes

$$\epsilon_k = -\log P\big(o_k^{\text{act}} = \hat{o}_k^{\text{act}} \bigm| q(s_k^{\text{type}}, s_k^{\text{stance}})\big) \,. \tag{1}$$

Equation (1) is the negative log posterior predictive likelihood assigned to the observed partner response under the current type/stance posterior. Equivalently, the predictive probability marginalizes over the agent's current beliefs about partner type and stance:

$$P\big(o_k^{\text{act}} = \hat{o}_k^{\text{act}} \mid q_k\big) = \sum_{s^{\text{type}}, s^{\text{stance}}} P\big(o_k^{\text{act}} = \hat{o}_k^{\text{act}} \mid s^{\text{type}}, s^{\text{stance}}\big)\, q_k(s^{\text{type}}, s^{\text{stance}}).$$

Partner response directly tests the type-by-stance model: a reliably defecting partner can produce low surprisal despite poor payoff, while an erratic cooperative partner can produce high surprisal despite favorable outcomes. The signal therefore tracks partner-response model fit rather than partner value.

The neutral baseline is the squared surprisal of a chance-level two-outcome partner-response prediction,

$$\sigma_0^2 = (-\log 0.5)^2.$$

Because the response modality is binary, probability 0.5 corresponds to predicting neither cooperation nor defection better than chance. Squaring places surprisal on a nonnegative error scale before signed affective charge is computed:

$$\phi_k = \alpha\big(\sigma_0^2 - \epsilon_k^2\big) \,, \tag{2}$$

where $\alpha \geq 0$ is a gain parameter. Positive ϕ_k means partner k's response was predicted better than chance; negative ϕ_k means it was predicted worse than chance.

The signed charge ϕ_k is then used as evidence over the partner-specific inverse-precision estimate β_k. Positive charge shifts posterior mass toward lower β_k, corresponding to higher policy precision; negative charge shifts posterior mass toward higher β_k, corresponding to lower policy precision. A persistence prior smooths this update across rounds so that confidence changes gradually rather than resetting after each interaction. The full categorical update is given in Appendix B.

The agent then sets partner-specific policy precision from the posterior mean:

$$\gamma_k = \gamma_{\text{base}} \,/\, \mathbb{E}_{q(\beta_k)}[\beta_k], \tag{3}$$

where γ_{base} is the baseline policy precision. Because β_k is inverse precision, higher expected β_k yields broader, less committed policy inference, whereas lower expected β_k yields higher γ_k and sharper policy commitment.

The β_k posterior is maintained as an auxiliary precision tracker rather than as a hidden state inside the focal agent's generative model. This keeps the mechanism lightweight and experimentally isolable, but prevents the agent from forming prior expectations over its own future confidence states; this limitation is discussed in Section 4.

2.4 Cross-partner policy selection

In partner-choice regimes, policy inference compares candidate partner–investment policies across relationships. For each partner k, the set Π_k contains policies involving that partner, such as different investment levels or action sequences directed toward k. The corresponding partner model $\mathcal{M}_k$ assigns each candidate policy an unnormalized policy-evidence score $u_{k,\pi}$, where larger values indicate stronger posterior support under the sign convention used for softmax policy inference. Partner-specific precision γ_k should sharpen or flatten commitment among policies involving partner k, without by itself changing the average evidence for selecting that partner.

To apply γ_k locally, we separate each partner's mean policy evidence from the relative differences among that partner's policies. Let

$$\bar{u}_k = \frac{1}{|\Pi_k|} \sum_{\pi \in \Pi_k} u_{k,\pi}$$

be the mean policy evidence for partner k. The precision-adjusted score is then

$$\tilde{u}_{k,\pi} = \bar{u}_k + \gamma_k \left(u_{k,\pi} - \bar{u}_k\right), \tag{4}$$

where γ_k is the partner-specific policy precision from Eq. (3). The transformation preserves the mean evidence for partner k, while sharpening or flattening relative preferences among policies directed toward that partner. Higher γ_k amplifies within-partner differences, producing more decisive policy commitment; lower γ_k compresses them, producing more tentative commitment.

The final policy posterior is computed by applying a softmax over the transformed scores $\tilde{u}_{k,\pi}$ across all partners and candidate policies. This procedure does not change the per-partner generative models; it only determines how partner-local precision enters cross-partner policy comparison.

2.5 Simulation setup

Simulations compare four precision-runtime variants: full partner-local precision, a shared-β ablation that pools evidence across partners while keeping per-partner

POMDP beliefs intact, a tracked-only ablation that updates $q(\beta_k)$ but fixes $\gamma_k = \gamma_{\text{base}}$, and a no-affect control that disables the β tracker entirely. Conditions include open graded investment, partner choice, abrupt betrayal, and precision-dynamics/profile diagnostics. Full condition definitions, episode lengths, and metrics are provided in Appendix C. Simulation code, analysis scripts, and experiment configurations will be released in a public repository upon acceptance.

3 Results

The experiments are organized around three claims about partner-local affective precision, followed by profile analyses of trust-calibration dynamics. Section 3.1 tests whether partner-local precision tracks partner-response predictability more strongly than realized payoff, and whether this relationship weakens when precision evidence is pooled across partners. Section 3.2 tests whether behavioral effects arise through the $\beta_k \to \gamma_k$ policy-precision pathway rather than improved partner-state inference. Sections 3.3 and 3.4 examine how this policy-commitment effect appears in partner choice and under abrupt social change. Section 3.5 then varies precision gain and prior model fitness to characterize temporal profiles of confidence revision.

3.1 Partner-local precision tracks predictability rather than realized payoff

Partner-local precision was more strongly associated with partner-response likelihood surprisal than with realized payoff. Because partner-choice episodes couple payoff, surprisal, and exposure, we computed active-encounter partial correlations controlling for encounter count (uncontrolled correlations in Appendix C.6). The precision–surprisal association remained strong ($r = -0.882$), while the precision–payoff association weakened substantially ($r = 0.130$). Negative signs indicate that higher partner-response likelihood surprisal corresponds to lower β_k-derived policy precision; Figure 1 plots absolute magnitudes for readability.

Pooling precision evidence across partners weakened this relationship-specific signal. In the shared-β ablation, per-partner beliefs remained separate but the precision tracker was updated from pooled partner evidence. The precision–surprisal association fell to $r = -0.380$ after controlling for encounter count, and the precision–payoff association remained weak ($r = 0.049$). Thus, the update rule alone is not sufficient: precision tracks partner-response model fit most clearly when confidence evidence remains tied to the partner who generated it.

Payoff over the same locality-probe analysis window did not favor the partner-local variant: cumulative payoff was 483.5 for partner-local precision, 517.6 for shared-β, and 542.1 for no-affect. The payoff panel uses the same active-encounter analysis window as the correlation readouts, rather than full-episode payoff. Thus, the strongest partner-local precision signal occurred without a corresponding payoff advantage, consistent with affective precision serving as a confidence-calibration signal rather than a reward-maximization mechanism.

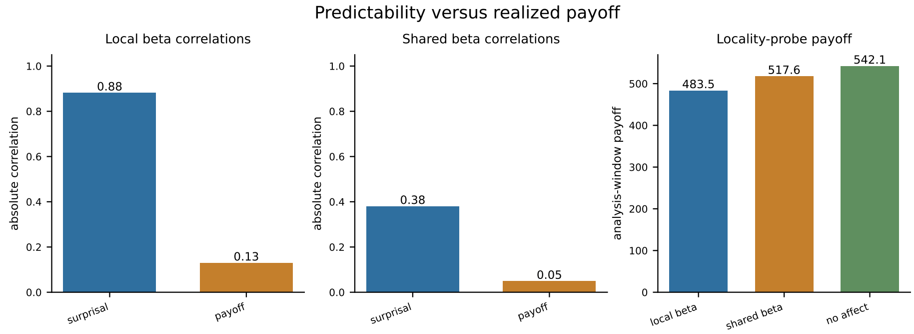

Fig. 1: Partner-local precision is more strongly associated with partner-response likelihood surprisal than with realized payoff. The correlation panels plot absolute active-encounter partial correlations; the payoff panel shows payoff over the same locality-probe analysis window. Pooling precision evidence across partners weakens the relationship between precision and partner-response likelihood surprisal, indicating that the signal depends on partner-local evidence rather than payoff alone.

3.2 Tracked-only ablation isolates policy commitment

To test whether partner-local affective precision changes behavior through policy deployment rather than partner-state inference, we used a tracked-only ablation. This variant updates $q(\beta_k)$ normally while holding $\gamma_k = \gamma_{\text{base}}$, preserving the precision estimate but preventing it from modulating policy selection.

The tracked-only ablation matched the no-affect baseline: both produced the same policy entropy (8.79 versus 8.79) and cumulative payoff (1864.2 versus 1864.2). Full precision modulation was the only variant that lowered entropy (8.59), with slightly lower cumulative payoff (1851.3). Thus, β_k matters behaviorally when it is allowed to modulate γ_k, producing sharper policy commitment. When that pathway is cut, the behavioral readouts match the no-affect baseline. Figure 2 shows the three-way comparison.

This supports the interpretation of affective precision as a calibration layer rather than an inference-improvement mechanism. The tracker does not directly improve partner-state inference; it regulates how strongly the agent commits to policies based on its current partner model. Whether sharper commitment improves payoff depends on the social context, not on precision alone.

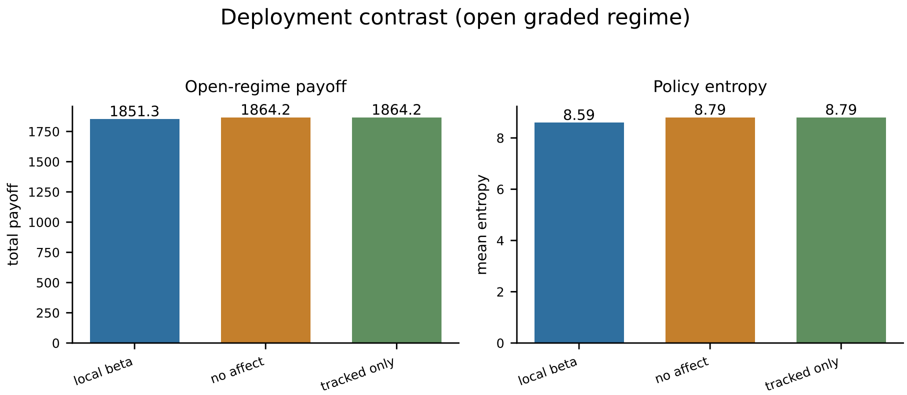

Fig. 2: Open graded regime: partner-local affective precision lowers policy entropy relative to no-affect and tracked-only controls without a corresponding payoff advantage. Tracked-only updates β_k but fixes γ_k; its readouts match the no-affect baseline, isolating the entropy effect to the $\beta_k \to \gamma_k$ policy-precision pathway.

3.3 Partner choice extends policy commitment to social allocation

Partner-local precision also affected policy commitment when the agent could choose whom to engage. In the partner-choice regime, each policy specifies both a partner and an investment level. Precision modulation lowered policy entropy relative to the no-affect control (3.99 versus 4.83), indicating that the $\beta_k \to \gamma_k$ pathway shapes partner–investment policies, not only investment levels within a fixed interaction.

The allocation shift was small and condition-specific: the partner-local agent selected the cooperator slightly more often than the no-affect control (36.6% versus 34.8%) and the exploiter slightly less often (13.8% versus 16.2%). This should not be read as a general cooperator-seeking bias. The main effect is sharper commitment under current partner models, not direct value tracking; aggregate payoff was nearly unchanged (393.6 versus 393.2).

Together, these results show that partner-local precision can reshape engagement without guaranteeing reward improvement. Whether sharper commitment helps or hurts depends on whether the social environment remains stable. The abrupt-betrayal condition tests this boundary directly.

3.4 Abrupt betrayal tests confidence under change

Accumulated confidence helps when it remains attached to a valid partner model, but can lag behind abrupt social change. The abrupt-betrayal condition tests this boundary: a previously cooperative partner switches stance at round 31, and the agent must adapt while carrying confidence built from the pre-switch relationship.

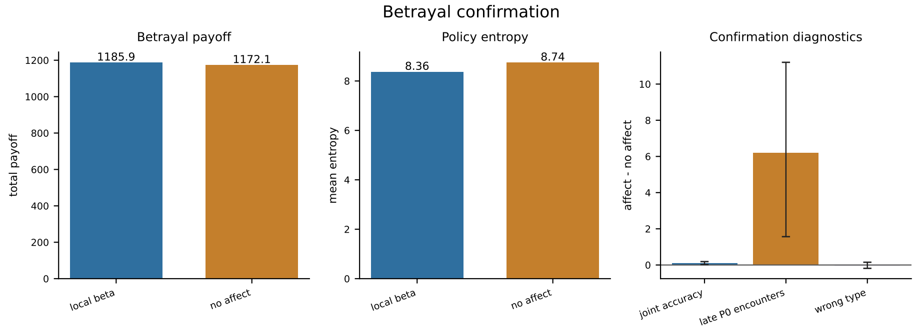

Fig. 3: Abrupt betrayal in the partner-choice regime. Partner-local affective precision lowers policy entropy relative to the no-affect control and increases joint partner-type accuracy, while the payoff difference remains uncertain. The result supports the interpretation of affective precision as a calibration mechanism whose deployment effects are reliable, but whose payoff consequences depend on post-change reallocation dynamics.

Precision modulation continued to affect deployment after the switch. Mean policy entropy was lower under partner-local precision than under the no-affect control (8.36 versus 8.74), with a bootstrap interval that did not cross zero (-0.62 to -0.14). Joint partner-type accuracy was also higher under precision modulation (0.372 versus 0.266; bootstrap interval 0.034 to 0.185), consistent with engagement being redirected rather than simply disrupted. Because the tracked-only ablation in Section 3.2 isolated the $\beta_k \to \gamma_k$ pathway, this accuracy difference is best interpreted as a downstream effect of changed engagement and sampling, not as direct improvement to the partner-state update.

Payoff showed a weaker and more variable pattern. The payoff difference was positive but uncertain (1185.9 versus 1172.1; bootstrap interval -25.2 to 53.2). This pattern is consistent with a calibration mechanism: confidence can keep policies decisive after a relationship changes, but reward depends on whether the agent redirects engagement quickly enough and toward useful alternatives. Figure 3 summarizes the post-switch readouts; the diagnostic panel reports raw affect-minus-no-affect differences in secondary measures that support the reallocation interpretation.

The betrayal result exposes a timing problem in social confidence: confidence built from reliable past evidence can remain behaviorally active after the relationship changes. We next vary precision gain and prior model fitness to examine how confidence-revision dynamics differ across agents.

3.5 Precision dynamics as computational profiles of social trust calibration

The betrayal result shows that confidence revision has temporal structure: confidence can be too weak, too slow, or too reactive. Varying precision gain α confirmed that gain controls the amplitude of β_k dynamics: in the betrayal regime, mean within-episode β_k range increased from 0.147 at $\alpha = 0.05$ to 1.218 at $\alpha = 8.0$. Payoff did not follow this monotonic pattern, indicating that faster confidence revision is not automatically better revision.

Crossing prior model fitness with gain produced distinct trust-calibration profiles. Low gain muted confidence dynamics and weakened policy commitment; high gain amplified revision but increased vulnerability to overcommitment after change; cautious priors suppressed premature confidence but also reduced exploitation of reliable partners. In the tested betrayal profiles, the default configuration achieved the highest payoff, while naive-stubborn, avoidant-rigid, and anxious-reactive variants expressed different failure modes: diffuse commitment, suppressed exploitation, and post-change overcommitment. Exact priors, gain values, and profile metrics are reported in Appendix D.

Trust-repair diagnostics further separated reengagement from restored confidence. Some variants reengaged a reverted partner without rebuilding the same confidence dynamics, while high-gain variants showed larger β_k swings without faster payoff recovery. The agent can return to a partner before it trusts them again. Extended profile and repair metrics are reported in Appendix D.

4 Discussion

The results support partner-local affective precision as a calibration mechanism for social policy commitment. Precision was defined from partner-response predictability rather than payoff, matched the no-affect baseline when the $\beta_k \to \gamma_k$ pathway was cut, and remained behaviorally active under abrupt change without guaranteeing reward improvement. Together, these findings support the paper's central distinction: social prediction and social commitment are related but distinct. A reliably cooperative partner and a reliably defecting partner can both generate low partner-response likelihood surprisal because both are predictable; conversely, a favorable but inconsistent partner can remain difficult to model. Partner-local affective precision therefore estimates how well the current partner model is working, then regulates how strongly that model is expressed in policy.

The tracked-only ablation clarifies where the mechanism acts. Updating β_k without allowing it to modulate γ_k left behavior unchanged relative to the no-affect baseline, while full precision modulation sharpened policy commitment. This supports the interpretation that partner-local affective precision does not improve partner-state inference directly; it changes how strongly current partner beliefs are deployed. It also clarifies the distinction from affective-precision accounts centered on policy-inference quantities: here, the model-fitness signal is partner-response likelihood surprisal under the current type–stance posterior.

The abrupt-betrayal condition shows the main limitation of this first calibration layer. Confidence built from reliable past evidence can remain behaviorally active after the relationship changes. In the present model, β_k is an auxiliary tracker, so the agent does not explicitly infer when its own confidence has become stale. A fuller architecture should add volatility or change detection that discounts accumulated confidence when prediction-error structure shifts, and should make partner-choice policies more exploratory after detected change. Embedding β_k as a hidden state would also allow the agent to form prior expectations over its own future confidence rather than only updating confidence retrospectively.

The profile analyses generate computational hypotheses about trust calibration, connecting to broader work on social learning, psychosis, and attachment [3, 1, 5]. Precision gain and prior model fitness shaped update amplitude, sensitivity to change, and reengagement after repair, but no single profile was uniformly best. High gain expanded confidence swings without monotonic payoff improvement; low gain stabilized behavior but slowed revision; cautious priors protected against premature confidence while suppressing confidence even toward reliable partners. These tradeoffs suggest empirical tests on trust-game time series: whether human behavior separates predictability from value, policy deployment from partner-state inference, and reengagement from restored confidence. A further extension is to replace scripted partners with reciprocal active-inference agents that maintain their own beliefs, policies, and precision estimates.

5 Conclusion

We formalize social affect as a relationship-specific metacognitive signal: an estimate of how confidently an agent should select policies from its current model of each partner. Implemented as partner-local affective precision, this signal updates from partner-response likelihood surprisal and modulates policy precision during selection. Across simulations in a graded multi-partner trust game, precision tracks predictability more closely than payoff, affects behavior through the $\beta_k \to \gamma_k$ policy-precision pathway rather than improved partner-state inference, and remains behaviorally active when relationships change.

These results support a distinction between social prediction and social commitment. An agent can infer what a partner is likely to do while separately estimating how strongly that inference should guide policy commitment. Precision gain and prior model fitness shape this calibration process, producing tradeoffs in confidence accumulation, betrayal adaptation, reengagement, and repair. Future empirical work can test whether human trust-game behavior shows similar distinctions between predictability and value, inference and policy commitment, and reengagement and restored confidence. Partner-local affective precision offers a computational account of social metacognition in which relationships differ not only in what an agent believes about them, but in how strongly those beliefs should guide policy selection.

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Appendix

A Full POMDP Specification

This appendix specifies the POMDP components used by the focal agent.

A.1 Hidden state factors and observations

As described in Section 2.2, the focal agent's generative model for each partner k maintains three hidden factors:

$$\begin{aligned} s_k^{\text{type}} &\in \{\text{cooperator, reciprocator,} \\ &\qquad \text{exploiter, random}\} && \text{(hidden behavioral type)} \quad (5) \\ s_k^{\text{stance}} &\in \{\text{trusting, neutral, hostile}\} && \text{(hidden stance)} \quad (6) \\ s^{\text{own}} &\in \mathcal{A} = \{0, 1, 2, 3, 4, 5\} && \text{(executed investment level)} \quad (7) \end{aligned}$$

The own-investment state is *deterministic bookkeeping*: the agent always knows its executed investment level, and this factor makes that knowledge available to the observation model. It is not a psychological hidden variable. The joint hidden state per partner spans $4 \times 3 \times 6 = 72$ configurations.

Observations are partner response and payoff:

$$\begin{aligned} o_k^{\text{act}} &\in \{\text{cooperate, defect}\} && \text{(partner response)} \quad (8) \\ o_k^{\text{pay}} &\in \{5, 6, 7, 8, 9, 10, 10.5, 11, 11.5, 12, 12.5\} && \text{(focal agent's realized payoff)} \quad (9) \end{aligned}$$

A.2 Partner-response likelihood

The core of the generative model is the *partner cooperation likelihood table*, which defines the probability the partner cooperates given type and stance:

Table 1: Cooperation likelihood $P(\text{cooperate} \mid s^{\text{type}}, s^{\text{stance}})$.

Partner type	Trusting	Neutral	Hostile
Cooperator	0.95	0.80	0.55
Reciprocator	0.90	0.70	0.30
Exploiter	0.70	0.35	0.10
Random	0.60	0.50	0.35

This table is the foundation of the likelihood modality A_k^{act}. Entries define

$$P\left(o_k^{\text{act}} = \text{cooperate} \mid s_k^{\text{type}}, s_k^{\text{stance}}\right), \quad (10)$$

with $P(o_k^{\text{act}} = \text{defect} \mid \cdot) = 1 - P(o_k^{\text{act}} = \text{cooperate} \mid \cdot)$. The modality is uniform over s^{own}: partner response depends only on inferred type and stance. This operationalizes what the agent considers a predictable partner—a cooperative reciprocator in a trusting stance has high predicted cooperation; an exploiter in a hostile stance has low predicted cooperation.

A.3 Payoff likelihood

Payoff enters as a second observation modality $A_k^{\text{pay}} = P(o_k^{\text{pay}} \mid s^{\text{own}}, s_k^{\text{type}}, s_k^{\text{stance}})$. For investment $a = s^{\text{own}}$, endowment $E = 10$, and multiplier $M = 3$, the deterministic payoff map is

$$\pi(a, o_k^{\text{act}}) = \begin{cases} E - a + \frac{Ma}{2}, & o_k^{\text{act}} = \text{cooperate}, \\ E - a, & o_k^{\text{act}} = \text{defect}. \end{cases} \tag{11}$$

Because the partner response is not directly controlled by the focal agent, the payoff likelihood marginalizes over the partner-response likelihood:

$$\begin{aligned} &P(o_k^{\text{pay}} = v \mid s^{\text{own}}, s_k^{\text{type}}, s_k^{\text{stance}}) \\ &= \sum_{o^{\text{act}} \in \{\text{cooperate}, \text{defect}\}} \mathbb{I}\big[\pi(s^{\text{own}}, o^{\text{act}}) = v\big]\, P(o_k^{\text{act}} = o^{\text{act}} \mid s_k^{\text{type}}, s_k^{\text{stance}}). \end{aligned} \tag{12}$$

This makes higher investment beneficial when cooperation is likely and costly when defection is likely, while keeping payoff preferences inside the generative model.

A.4 Transition dynamics

Type drift (B_k^{type}). Partner type is uncontrollable and drifts slowly:

$$P(s_k^{\text{type}\prime} = j \mid s_k^{\text{type}} = i) = \begin{cases} 1 - p_{\text{switch}} & j = i, \\ p_{\text{switch}}/(K_{\text{type}} - 1) & j \neq i, \end{cases} \tag{13}$$

with $K_{\text{type}} = 4$ and default $p_{\text{switch}} = 0.05$. Scheduled betrayal scenarios set $p_{\text{switch}} = 0$.

Stance dynamics (B_k^{stance}). Stance transitions depend on the focal agent's graded investment a by interpolating between a high-investment/trust-building endpoint and a low-investment/withholding endpoint:

$$\rho(a) = \frac{a}{|\mathcal{A}| - 1} = \frac{a}{5}, \qquad B^{\text{stance}}(a) = \rho(a) B^{\text{high}} + (1 - \rho(a)) B^{\text{low}}. \tag{14}$$

High investment shifts dynamics toward trust building; low investment or withholding shifts dynamics toward trust damage; recovery from hostility is slower than collapse.

Table 2: Endpoint stance transition dynamics. The focal agent's graded investment interpolates between B^{high} and B^{low} according to Eq. (14).

High investment / trust-building endpoint B^{high}			
To \ From	Trusting	Neutral	Hostile
Trusting	0.90	0.30	0.05
Neutral	0.10	0.60	0.35
Hostile	0.00	0.10	0.60
Low investment / withholding endpoint B^{low}			
To \ From	Trusting	Neutral	Hostile
Trusting	0.10	0.05	0.02
Neutral	0.50	0.35	0.18
Hostile	0.40	0.60	0.80

These dynamics instantiate the asymmetry of trust: sustained high investment gradually builds trust, while sustained withholding rapidly damages trust; recovery from hostility is slow.

Own-investment bookkeeping (B^{own}). The own-investment factor updates deterministically from the executed investment level:

$$P(s^{\text{own}\prime} = a' \mid s^{\text{own}}, a) = \mathbb{I}[a' = a]. \tag{15}$$

This factor is not a psychological hidden variable; it makes the agent's executed investment level available to the payoff likelihood in standard factorized form.

A.5 Preferences, priors, and policy priors

Observation preferences (C_k). There is no direct preference over partner responses:

$$C_k^{\text{act}} = [0,\ 0] \quad \text{over } \{\text{cooperate}, \text{defect}\}. \tag{16}$$

Payoff preferences ascend over the graded payoff support generated by Eq. (11):

$$C_k^{\text{pay}} = \text{log\_softmax}\left(\frac{[5, 6, 7, 8, 9, 10, 10.5, 11, 11.5, 12, 12.5]}{\tau}\right), \tag{17}$$

where τ is a temperature parameter controlling preference sharpness. Instrumental motivation therefore enters through payoff observations and multi-step planning rather than direct preferences over partner responses.

Initial-state priors (D_k). Type, stance, and own-investment priors are

$$D_k^{\text{type}} = \left[\tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{4}\right], \tag{18}$$

$$D_k^{\text{stance}} = [0.2,\ 0.6,\ 0.2], \tag{19}$$

$$D^{\text{own}} = \text{Unif}(\{0, 1, 2, 3, 4, 5\}). \tag{20}$$

Policy priors (E_k). Policy priors are uniform over admissible graded investment sequences at the configured planning horizon. In partner-choice settings, partner selection is handled outside the per-partner POMDP by evaluating partner–investment policy branches with precision-adjusted policy scores before executing the chosen investment.

A.6 Partner implementation

Partner implementation follows Section 2.1. Four environment-side partners maintain hidden type and stance and sample cooperate-or-defect responses from the cooperation likelihood in Table 1, undergoing investment-dependent stance transitions per Table 2. Partners do not perform variational inference or maintain affective precision estimates, isolating the focal agent's precision mechanism before extending to reciprocal active-inference partners.

B Affective Precision Update and Policy Selection

This appendix gives the β-update rule, policy-precision mapping, and partner-choice policy-selection procedure.

B.1 Categorical beta support

Each partner maintains a categorical posterior $q(\beta_k)$ over discrete inverse-precision levels $\{0.5, 0.67, 1.0, 1.5, 2.0\}$, updated each round from affective charge ϕ_k. The categorical representation keeps the inverse-precision convention explicit, bounds the precision dynamics, and allows the update to be analyzed as movement over interpretable confidence states.

B.2 Persistence prior

The update proceeds in three steps. First, a persistence prior smooths the previous posterior via a tridiagonal transition matrix:

$$p(\beta_k) \leftarrow T \cdot q_{\text{prev}}(\beta_k), \tag{21}$$

where T is tridiagonal with persistence parameter 0.8 and reflecting boundaries, encoding the prior expectation that precision evolves slowly.

B.3 Charge-based pseudo-likelihood

Second, a charge-based pseudo-likelihood maps positive charge toward high policy precision (low β) and negative charge toward low policy precision (high β):

$$\log \ell(\beta_\ell) = \phi_k \cdot \frac{1}{\beta_\ell}, \tag{22}$$

where β_ℓ ranges over the five support points. This update satisfies the intended monotonicity: positive charge reinforces lower β, negative charge reinforces higher β, and the effect scales with effective precision at each support point.

B.4 Posterior update and policy precision

Third, the posterior is normalized:

$$q(\beta_k) \propto \exp(\log \ell(\beta_\ell)) \cdot p(\beta_k). \tag{23}$$

Policy precision is then set from the posterior mean as given in Eq. (3) of the main text. The quantity β_k is inverse precision, so higher posterior mean β_k lowers policy precision γ_k, while lower posterior mean β_k raises γ_k. The β_k posterior is maintained as an external auxiliary tracker rather than a proper hidden state in the generative model, which simplifies implementation at the cost of losing the agent's ability to form prior expectations over its own future affective states (discussed in Section 4).

B.5 Cross-partner policy selection

In partner-choice regimes, partner-specific precision sharpens or flattens policy commitment within a relationship while preserving the mean cross-partner evidence for engaging that partner. The transformation in Eq. (4) applies γ_k to deviations from the partner-specific policy mean, rather than to raw policy scores. The agent samples or selects policies from the combined set of transformed scores across all partners and policies. This lets γ_k control commitment among policies involving partner k without changing the average evidence for approaching partner k during cross-partner comparison.

B.6 Ablation variants

The four precision-runtime variants are defined in Section 2.5. For completeness: the no-affect control disables the β tracker entirely and fixes $\gamma_k = \gamma_{\text{base}}$; the tracked-only ablation updates $q(\beta_k)$ normally but fixes $\gamma_k = \gamma_{\text{base}}$, allowing the precision estimate to be analyzed without influencing policy selection; the default partner-local variant applies the full update pipeline described in B.1–B.4 and sets γ_k from each partner's posterior mean; and the shared-β ablation maintains a single pooled β state updated from all partner evidence while keeping per-partner POMDP beliefs intact, with all partner branches inheriting precision from the shared expected β.

C Simulation Protocols and Metrics

This appendix summarizes simulation conditions, replication counts, schedules, and metrics.

C.1 Simulation conditions

All reported experiments use one focal active-inference agent with four partners ($K = 4$), per-partner generative models, graded investment actions, and environment-side partner policies (Appendix A). Conditions differ by assignment mode, scheduled partner changes, and precision/profile manipulations within the graded payoff regime. In conditions without explicit partner choice, the interaction partner is assigned according to the configured protocol; in partner-choice conditions, the agent selects the partner before selecting an investment level.

Open graded decision regime. The graded regime uses a discretized investment level while keeping the same hidden type–stance structure. This is the primary deployment-contrast setting (Section 3.2): affect, no-affect, and tracked-only variants are compared on policy entropy and cumulative payoff.

Predictability-over-payoff locality probe. A 100-round partner-choice graded environment with partner-local, shared-β, and no-affect variants tests whether β_k-derived policy precision tracks partner-response likelihood surprisal more strongly than realized payoff (Section 3.1). The payoff panel reports the same active-encounter analysis window used for the locality-probe correlation readouts.

Partner-choice regime. The agent first selects a partner, then selects an investment level toward that partner each round. With four partners and six graded investment levels, the one-step partner-choice setting can equivalently be represented as $4 \times 6 = 24$ partner–investment branches. Partner selection is where engagement reorganization and social-allocation readouts are defined (Section 3.3).

Abrupt betrayal. A partner-choice graded environment with scheduled stance betrayal on a previously cooperative partner tests post-switch reallocation, entropy, payoff, and joint accuracy (Section 3.4).

Precision-dynamics/profile variation. Computational profile variants vary β_k amplitude and stability by manipulating precision gain α and prior model fitness in open graded, betrayal, and partner-choice environments (Section 3.5; Appendix D).

Forgiveness. The trust-repair protocol extends abrupt betrayal by allowing the betrayed partner to return to a cooperative/trusting state, testing reengagement and confidence recovery after restored reliability (Section 3.5).

C.2 Seed counts and episode lengths

Unless otherwise stated in the main text, episode length is 200 rounds. The abrupt-betrayal condition uses 120 rounds because post-switch readouts dominate interpretation. Profile-scale tables use 20 seeds per condition. Central behavioral comparisons use 30 seeds per condition. Exploratory diagnostic runs used three seeds per variant and were used only for implementation validation; all main-text claims are based on the confirmation-scale or profile-scale runs reported in the relevant sections.

Representative configured scales:

Table 3: Representative episode lengths and replication counts by experiment family.

Experiment family	Setting	Rounds	Seeds
Open deployment	graded, random/open choice	200	30
Predictability-over-payoff locality probe	graded partner choice	100	30
Partner allocation	partner choice	200	30
Abrupt betrayal	scheduled partner-choice betrayal	120	30
Precision perturbation	graded perturbations	200	30
Profile program	sweeps, factorial, forgiveness	200	20

Supplementary materials. The supplementary repository contains the simulation environments, precision-runtime variants, experiment configuration files, analysis scripts, and figure-generation code used for the reported experiments.

C.3 Betrayal schedule

The canonical abrupt-betrayal protocol (`betrayal_choice`) uses graded payoffs, agent-chosen partners, four partners with fixed initial types and stances, and no stochastic type switching ($p_{\text{switch}} = 0$). Partner P0 begins as a cooperator in a trusting stance. At round 31, P0 undergoes a scheduled stance switch to hostile while other partners retain their initial configurations. Primary readouts are post-switch cumulative payoff, policy entropy, partner reallocation, and joint partner-type accuracy over the remaining episode.

The profile-scale betrayal environments used in the alpha sweep and prior-gain factorial apply the same single-switch structure within 200-round episodes for recovery-time and quadrant metrics. The forgiveness profile extends this schedule: rounds 1–80 cooperative baseline, rounds 81–120 betrayal on P0, rounds 121–200 reversion to cooperative/trusting on P0.

C.4 Partner-choice allocation metrics

Partner-choice readouts quantify how precision reshapes *who* the agent engages, not only how it acts toward a fixed partner.

Selection share. Per-partner selection rate is the fraction of rounds on which each partner is chosen. Partner-choice reallocation is summarized qualitatively in Section 3.3.

Selection concentration. The Gini coefficient of the partner-selection distribution summarizes how concentrated engagement is across the social context (used in the profile alpha sweep and prior-gain factorial).

Post-betrayal allocation. Post-switch windows report reallocation away from the switched partner and toward alternatives, including post-betrayal P0 selection and high-investment rates in profile analyses.

Policy entropy under choice. Mean q_π entropy in partner-choice regimes captures how sharply the agent commits to partner–investment policies (reported comparatively for affect versus no-affect controls).

C.5 Entropy, payoff, and accuracy readouts

Total payoff. Cumulative focal-agent payoff over the episode (or over defined post-switch windows for betrayal analyses).

Policy entropy. Mean q_π entropy (‘q_pi_entropy‘ in logged outputs) measures policy-commitment sharpness; lower entropy indicates sharper policy selection under precision modulation.

Joint partner-type accuracy. Mean joint correctness of inferred partner type (‘mean_joint_accuracy‘) summarizes epistemic state inference; it is reported separately from payoff because precision modulation can change deployment and accuracy independently of cumulative reward.

β_k **range.** Unless otherwise specified, β_k range denotes the mean within-episode range of the posterior mean $\mathbb{E}_q[\beta_k]$, averaged across relevant partners and seeds. It is an observed precision-dynamics amplitude metric, not the fixed support of the categorical β_k variable.

Profile diagnostics. Investment churn is the fraction of adjacent rounds in which the selected investment level changes. Early exploiter investment is the fraction of rounds 1–30 in which the selected partner is an exploiter and the investment level is at least half of the available action range. Trust asymmetry is the ratio of withdrawal latency after the first defection to initial high-investment approach latency.

Betrayal timing metrics. Recovery rounds and related latency readouts measure rounds after a switch until selection or payoff returns toward baseline thresholds (profile alpha-sweep, prior-gain factorial, and betrayal analyses).

C.6 Locality-probe correlation readouts

For the predictability-over-payoff locality probe (Section 3.1), correlations are computed at the partner–seed unit level ($n = 120$ units across 30 seeds). Main-text claims use active-encounter partial correlations that control for encounter count, because partner-choice episodes couple payoff, surprisal, and exposure. Uncontrolled Pearson correlations are reported here for completeness.

Table 4: Locality-probe correlations by precision-runtime variant.

Variant / association	Raw r	Partial r
Partner-local precision		
Precision–surprisal	-0.949	-0.882
Precision–payoff	-0.748	0.130
Shared-β ablation		
Precision–surprisal	-0.555	-0.380
Precision–payoff	-0.447	0.049

Negative precision–surprisal associations indicate that higher partner-response likelihood surprisal corresponds to lower β_k-derived policy precision. Figure 1 plots absolute partial-correlation magnitudes for the partner-local and shared-β variants.

C.7 Replication summary

Central behavioral comparisons in Section 3 use 30 seeds per condition. Profile-scale analyses use 20 seeds per condition.

D Extended Profile and Robustness Results

This appendix reports extended parameter sweeps, profile analyses, and robustness diagnostics that support the main-text results.

D.1 Precision-dynamics parameter variation

We first vary the amplitude and stability of β_k dynamics to test how precision dynamics affect policy commitment. These variants are computational controls rather than diagnostic categories.

Table 5: Precision-dynamics parameter variation in the graded decision regime.

Variant	Mean β_k range	Total payoff	Policy entropy
affect	1.32	1851.3	8.59
low-gain	0.24	1840.5	8.9
high-gain	1.48	1859.0	8.1
cautious-prior	1.48	1877.7	8.4

Flattening the precision signal produces the clearest behavioral change. The low-gain variant compresses β_k range to 0.24 and raises entropy to 8.9, near the no-affect and tracked-only baselines. Expanded-dynamics variants produce wider β_k movement, but payoff does not follow range monotonically. The parameter sweeps below therefore vary gain and prior structure directly.

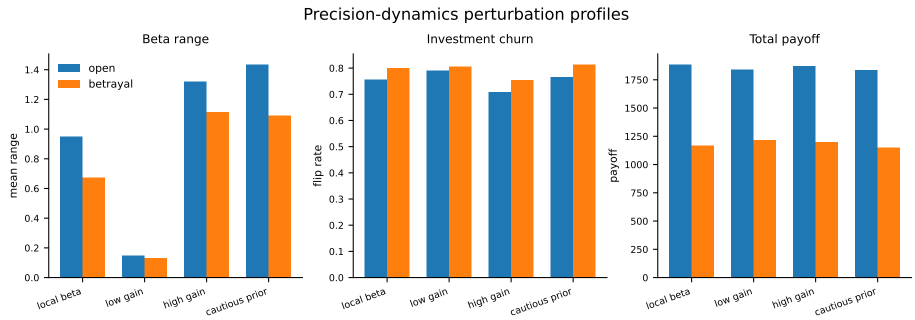

Fig. 4: Precision-dynamics perturbations in the graded decision regime. Low gain compresses β_k dynamics; wider β_k movement does not produce monotonic payoff gains. Entropy values for the same variants are reported in Table 5.

D.2 Alpha sweep

Sweeping α across $\{0.05, 0.1, 0.3, 0.5, 1.0, 2.0, 4.0, 8.0\}$ tests whether precision gain controls reactive social confidence across metrics. β_k movement range increases monotonically with α in the betrayal regime:

Table 6: β_k range by precision-gain α (betrayal regime, 20 seeds).

α	Mean β_k range
0.05	0.147
0.1	0.182
0.3	0.326
0.5	0.416
1.0	0.580
2.0	0.862
4.0	1.051
8.0	1.218

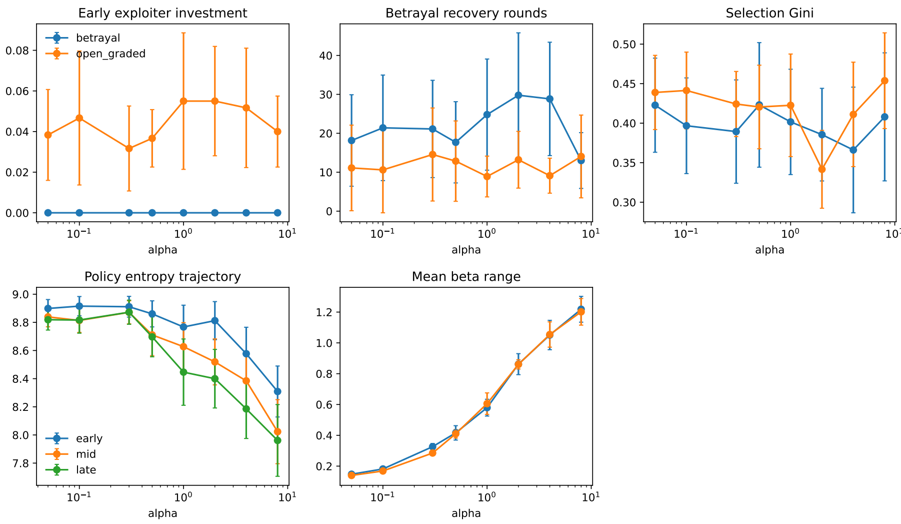

Fig. 5: Precision-gain sweep across open graded and betrayal environments (20 seeds). Higher α expands β_k range and reshapes engagement readouts without monotonic payoff ranking.

D.3 Gain-prior profile quadrants

Crossing prior model fitness with precision gain defines the profile variants used in Section 3.5. "Naive" and "cautious" refer to the initial prior over $q(\beta_k)$ on support $\{0.5, 0.67, 1.0, 1.5, 2.0\}$: naive uses $[0.40, 0.40, 0.15, 0.04, 0.01]$, while cautious uses $[0.01, 0.04, 0.15, 0.40, 0.40]$. Low and high gain refer to $\alpha = 0.1$ and $\alpha = 3.0$, respectively, with the default agent using $\alpha = 3.0$ and centered initialization at $\beta_k = 1.0$. β_k range denotes the mean within-episode range of the inverse-precision estimate. Table 7 gives key metrics.

Table 7: Behavioral metrics by profile (betrayal environment, 20 seeds). Default agent shown for reference.

Profile	Payoff	Recovery rounds	Gini	β_k range	Trust asymmetry
Anxious-reactive	1932	26.5	0.338	1.084	0.98
Hypervigilant	1951	20.1	0.423	0.892	1.65
Naive-stubborn	1894	21.3	0.304	0.364	1.19
Avoidant-rigid	1889	14.9	0.357	0.322	1.21
Default	1991	23.0	0.423	0.893	1.68

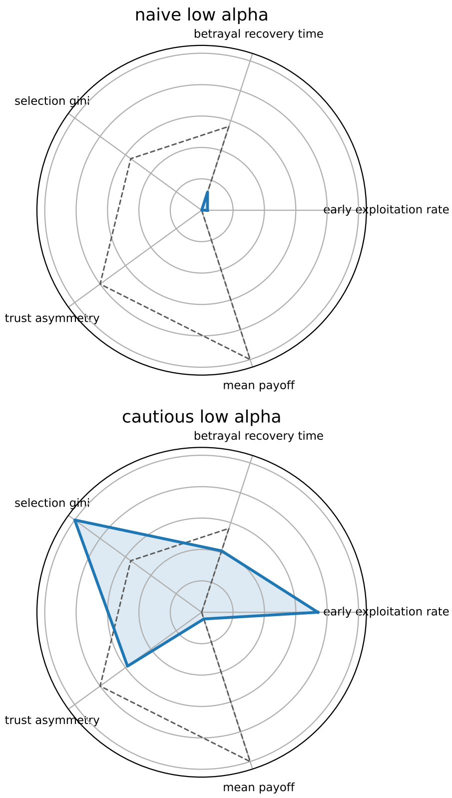

Fig. 6: Gain-prior profile quadrants from the prior-by-α factorial (betrayal environment, 20 seeds).

D.4 Forgiveness and trust repair

The forgiveness paradigm uses a 200-round partner-choice episode: cooperative baseline (rounds 1–80), betrayal on P0 (rounds 81–120), then reversion to cooperative/trusting behavior (rounds 121–200). Readouts include reengagement latency, post-return selection, payoff recovery, and β_k recovery trajectories for the reverted partner.

Table 8: Forgiveness and trust-repair metrics by precision profile (20 seeds). Reengagement is the post-repair P0 selection rate; payoff recovery compares late post-repair payoff with the late pre-betrayal baseline.

Variant	Reengagement	Payoff recovery	Latency	β_k range
cautious-high-α	0.518	0.996	5.5	0.971
cautious-low-α	0.630	1.014	1.5	0.287
default	0.475	1.019	3.0	1.012
naive-high-α	0.560	1.005	2.2	1.100
naive-low-α	0.527	1.004	4.6	0.325
no-affect	0.593	1.033	1.7	–

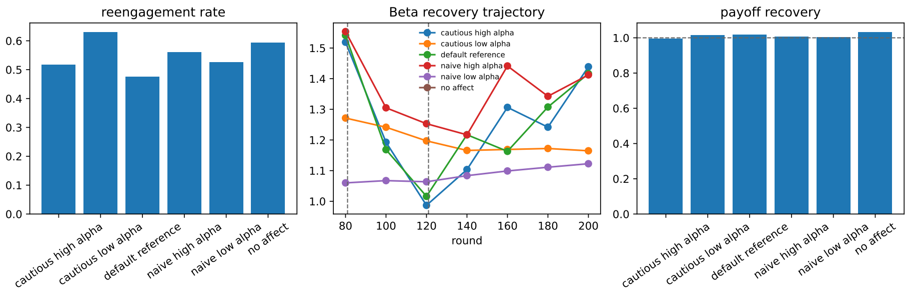

Fig. 7: Forgiveness/trust-repair diagnostic from the 20-seed profile analyses. Reengagement and payoff recovery do not collapse into a single monotonic gain effect: no-affect and cautious-low-α profiles reengage most often, while payoff recovery remains near baseline across profiles.